BANTU INFOTECH

(B.iT)

INV-2026-010 • SITA Security Platform & FMCG Ingestion Data Lake Backend Architecture

Period: 23 September 2026 – 22 October 2026 | Date: 22 October 2026

PREPARED BY

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CLIENT

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Project: SITA Security Intelligence Platform (Addendum B, AppScan SAST/DAST Remediation, AIMI AI Layer) & FMCG Enterprise Ingestion, Data Lake & Predictive Analytics Backend Architecture
Contract: Monthly Retainer under Independent Contractor Agreement (01 May 2026 – 30 April 2028 per ADD-2026-002)
Period: 23 September 2026 – 22 October 2026 | **Date:** 22 October 2026 | **Invoice No:** INV-2026-010
Contact: admin@hosi.co.za | **Monthly Retainer:** R60,000.00

PART 1 – FLIGHT PLANNING & ADDENDUM B COMPRESSION (23 SEP – 07 OCT) — 18 HRS

#	DELIVERABLE / DESCRIPTION	HRS
1.1	**Unified Flight Plan Addendum B & Execution Decomposition** • Re-engineered flight plan compression from 12-day to 9-day execution window (Plan v4) • Authored HOSI-UNIFIED_FLIGHT_PLAN_ADDENDUM_B_G2_9DAY_COMPRESSION.xlsx decomposing 39 discrete engineering tasks (T01–T39, 14.25 PD) • Established strict boundary verification: separated prompt authoring from verification, with verifier containment test assertions • Authored scratch/build_9day_addendum.py (1,715 lines) validating resource capacity, float, and gate milestones	9
1.2	**Architecture Decision Records & Evidence Pack Framework** • Authored comprehensive governance suite: docs/AI_INTEGRATION_PLAN_MASTER.md (660 lines) and 01_G2_POC_Plan_2026-10-05_to_2026-10-15.md • Authored 02_G2_POC_Decision_Papers_D1-D8.md, 05_G2_Agent_Architecture_Assessment.md, and 06_G2_Architecture_Decision_Record.md • Compiled 08_Evidence_Pack.md (DEL-19): non-interference proof, verification-independence proof, latency budget register, and acceptance checklist • Implemented cryptographic ECDSA P-256 PKI Attestation Keypair engine (backend/app/security_pki.py) for attestation signing and verification	9
Part 1 Subtotal		**18**

PART 2 – MULTI-TENANCY HARDENING & GOVERNANCE EVIDENCE (07 – 12 OCT)		18 HRS
#	DELIVERABLE / DESCRIPTION	HRS
2.1	**Regulatory Alignment & Security Mapping Matrix** • Authored docs/B3_POPIA_Compliance_Assessment.md and docs/B4_POPIA_Remediation_Plan.md establishing zero-log PII guardrails • Formulated compliance traceability documentation: C1_Requirements_Traceability_Matrix.md, C2_Framework_Security_Mapping.md, and C3_CISO_Four_Tier_Role_Mapping.md • Authored C4_Customer_Outcomes_Persona_Matrix.md, A1_Automated_Test_Report_Pack.md, and A2_Master_Test_Registry.md (2,347 lines) mapping 150+ automated test verifications • Developed scripts/export_master_registry.py with defusedxml and canonical path validation	10
2.2	**Infrastructure Hardening & Reverse Proxy Modernization** • Replaced generic Nginx config with hardened infrastructure/nginx/sita-http.conf • Pinned Nginx upstream frontend port to 8080 and exposed secured backend API on port 8000 in docker-compose.remote.yml • Implemented server token suppression, HR-backed user provisioning enforcement, and configurable environment controls (SEED_DEMO_USERS_ENABLED, PII_ENCRYPTION_KEY)	8
Part 2 Subtotal		**18**

PART 3 – FMCG DISTRIBUTED DATA LAKE & INGESTION BACKEND ARCHITECTURE (07 – 13 OCT)		22 HRS
#	DELIVERABLE / DESCRIPTION	HRS
3.1	**High-Throughput Streaming Ingestion Pipeline & Enterprise Reliability** • Architected asynchronous ingestion pipeline (app/ingestion/fmgc_pipeline.py) streaming 12,500+ ERP order records from raw multi-sheet workbooks into PostgreSQL lake • Engineered enterprise fault-tolerance: Three-state Circuit Breaker (circuit_breaker.py), Token-Bucket rate limiter (token_bucket.py), and Exponential Backoff Retry engine (retry.py) • Implemented Dead-Letter Queue (dlq.py) isolating invalid/malformed records into staging.rejected_records with failure payloads and ingestion audit trails • Developed dual-bus event architecture supporting Redis Streams (STREAM_FMGC_RAW) with zero-config local memory bus fallback (local_bus.py)	11
3.2	**PostgreSQL Lake Engine, Normalisation Rules & Predictive Analytics Models** • Designed multi-schema Postgres lake (fmgc, staging, warehouse) with chunked transactional upsert engine (fmgc_writer.py, 500-record batches) • Built declarative normalisation engine (fmgc_normaliser.py, field_mapper_fmgc.yaml) handling Excel sentinel dates (1900-01-01 to NaT), whitespace trimming, and net/gross lost value logic • Developed predictive analytics suite (predictive.py): Logistic Regression delay-risk classifier (ROC/PR tuned), Random Forest delivery ageing regressor with multi-horizon SLA breach probabilities (7/14/30/45 days), and additive Holt-Winters intake demand forecaster • Built high-performance FastAPI service (:13000, /api/v1/fmgc/*) with dual-TTL fingerprint-keyed memory cache (cache.py) and automated executive HTML report generation (reports.py)	11

#	DELIVERABLE / DESCRIPTION	HRS
Part 3 Subtotal		**22**

PART 4 – HCL APPSCAN DAST & SAST VULNERABILITY REMEDIATION (12 – 14 OCT) — 22 HRS

#	DELIVERABLE / DESCRIPTION	HRS
4.1	**HCL AppScan DAST Dynamic Vulnerability Remediation & Proxy Hardening** • Remediated HCL AppScan DAST assessment findings across production target (Oct 13-14 cycles) • Enforced strict algorithm verification in JWT decoding, explicitly rejecting alg: none exploits • Implemented mass assignment guards preventing privilege escalation and per-tenant/per-IP rate limiting middleware • Upgraded Nginx to 1.27.4-alpine (resolving CVE-2025-23419), purged unsafe-inline/unsafe-eval from CSP, and remediated BFLA (CWE-284) on metrics endpoints • Authored docs/SITA_DAST_VAPT_Report_Oct2026.docx and docs/SITA_VAPT_Remediation_Report_Oct2026.docx	**11**
4.2	**SAST Static Code Remediation & DefusedXML/Cryptography Hardening (49 Findings)** • Resolved 49 HCL AppScan SAST static security findings across backend codebase • Upgraded alert deduplication hashing from MD5 to collision-resistant SHA-256 (alert_dedup.py) and adopted secrets.SystemRandom for backoff jitter (CWE-327) • Hardened backend Dockerfile: pinned python:3.12.7-slim, established non-root appuser:appgroup (CWE-266), dropped root capabilities, and enforced read-only root filesystems • Mitigated XML External Entity (XXE, CWE-91) via defusedxml, eliminated Path Traversal (CWE-73) via canonical Path.resolve(), and purged hardcoded credential heuristics • Authored docs/SITA_SAST_Remediation_Report.docx, docs/SAST_REMEDIATION_REPORT.md, and docs/SITA_SAST_Comparative_Analysis_Report_Oct2026.docx	**11**
Part 4 Subtotal		**22**

PART 5 – AIMI AI INTELLIGENCE & 5 AVP ENDPOINTS (14 – 15 OCT) — 24 HRS

#	DELIVERABLE / DESCRIPTION	HRS
5.1	**AIMI VAPT Log Intelligence Engine** • Architected and built backend/app/processing/intel/vapt.py: automated log intelligence parser performing severity clustering, pattern extraction, and threat narrative generation • Implemented end-to-end unit test suite (backend/tests/test_vapt_log_intel.py) covering 8 deterministic scenarios • Built frontend API client (frontend/src/api/intel.ts) and interactive UI panel AimiVaptLogInterpreterPanel within frontend/src/pages/AiIntelPage.tsx • Gated endpoints POST /intel/vapt/interpret-logs and GET /intel/vapt/baseline-status with tenant and RBAC controls	**8**
5.2	**Five Core AI Value Proposition (AVP) Endpoints** • Developed and integrated 5 core AI Value Proposition residual modules (backend/app/processing/intel/):	**10**

#	DELIVERABLE / DESCRIPTION	HRS
	• AVP-2 (Detection Content): author_rules.py (POST /intel/attack/author-rule) proposing git-diff alert rules without automated merges • AVP-3 (Threat Enrichment): campaign.py (POST /intel/threat/attribute) extracting ATT&CK/CVE techniques against estate exposure • AVP-5 (Legal Guidance): popia.py (POST /intel/popia/dossier) generating POPIA Section 22 breach notification evidence packages • AVP-8 (Causal Analysis): causal.py (POST /intel/risk/causal) hypothesizing root causes and mapping dominant telemetry to ATT&CK • AVP-11 (Behavioral Auditing): behaviour.py (POST /intel/audit/explain) identifying temporal burst anomalies (RAPID_AUTHORITY_SWEEP, CROSS_SCOPE_PIVOT)	
5.3	**Threat Attribution & Advisory Review Boundary Invariants** • Enforced strict architectural safety invariant across all AI intelligence endpoints: advisory_only=True • Implemented zero automated database writes and mandatory human analyst sign-off workflows • Validated cross-tenant isolation and strict schema conformance across all five AVP modules	6
Part 5 Subtotal		**24**

PART 6 – G2 AI REPORT GENERATION ENGINE & COPILOT (15 – 18 OCT) — 18 HRS

#	DELIVERABLE / DESCRIPTION	HRS
6.1	**G2 Two-Gate Verification Engine** • Built G2 report generation architecture (backend/app/ai/): • Gate 1 (Pre-generation Scope Filter): strict tenancy validation and role permission enforcement (backend/app/ai/scope.py, permissions.py) • Gate 2 (Post-generation Containment Verifier): automated assertion engine verifying zero hallucination and tenant boundary compliance (backend/app/ai/verify.py) • Developed context assembly pipeline (backend/app/ai/context.py) binding AI models directly to ingested tenant telemetry logs • Exposed secured POST /api/v1/reports-ai endpoint with full unit test coverage (backend/tests/test_reports_ai.py)	10
6.2	**Frontend Interactive Grounded Copilot & Report Studio** • Developed frontend/src/components/ai/AiReportStudio.tsx (1,021 lines): interactive studio featuring streaming report preview, tenant selection, two-gate status indicators, and export actions • Built frontend/src/api/reportsAi.ts typed client and role-aware navigation integration	8
Part 6 Subtotal		**18**

PART 7 – AIMI MULTI-PROVIDER GATEWAY & DBSEC AI ENGINE (19 – 21 OCT) — 22 HRS

#	DELIVERABLE / DESCRIPTION	HRS
7.1	**AIMI Multi-Provider Gateway Abstraction** • Implemented multi-provider abstraction gateway (backend/app/ai/aimi/gateway.py, 375 lines) supporting Ollama, OpenAI, and Anthropic backends with automatic failover	6

#	DELIVERABLE / DESCRIPTION	HRS
	• Authored unit test suite backend/tests/test_aimi_integration.py (425 lines) verifying deterministic provider routing and error recovery	
7.2	**Real-time Contextual Enrichment & POPIA Zero-Log Sanitizer** • Built real-time contextual enrichment engine (enrichment.py, 245 lines) augmenting threat prompts with localized tenant telemetry • Implemented POPIA zero-log PII redaction sanitizer (redaction.py, 145 lines) scrubbing RSA ID numbers, phone numbers, and employee credentials before external inference	5
7.3	**Database Security (DbSec) AI Alerts Engine & React UI** • Implemented database security anomaly detection engine (backend/app/processing/intel/dbsec.py, 555 lines) detecting exfiltration, privilege abuse, and anomalous queries • Authored test suite backend/tests/test_dbsec_ai_alerts.py (265 lines) • Developed frontend DbSecAiAlertsPanel.tsx and React hook useDbSecAiAlerts.ts	6
7.4	**Stakeholder Value Documentation & OpenAPI 3.1 Technical Artefacts** • Generated executive value reports: generate_aimi_stakeholder_report.py -> docs/SITA_AIMI_AI_Integration_Business_Value_Proposition.docx, generate_security_progress_report.py -> docs/SITA_Security_Assessment_Progress_Report_Oct2026.docx, generate_truth_remediation_report.py -> docs/SITA_Security_Remediation_Executive_Truth_Report_Oct2026.docx • Compiled complete OpenAPI 3.1 specifications (openapi.json, openapi.yaml) and ingestion documentation (INGESTION_PIPELINE_EXPLAINED.md)	5
Part 7 Subtotal		**22**

PART 8 – C4 ARCHITECTURE VIEWER, DECKS & EXECUTIVE BRIEFS (21 – 22 OCT) **16 HRS**

#	DELIVERABLE / DESCRIPTION	HRS
8.1	**Interactive C4 Architecture Viewer & Handover Decks** • Enhanced interactive architecture viewer (architecture.html, 2,116 lines) incorporating comprehensive AI integration layers, inspection modals, and zero-log markers • Updated handover_presentation.html and presentation.html incorporating 4-layer AI control model, live demo flows, and granular RBAC-within-tenancy matrix	8
8.2	**Executive Briefs Generation, UI Polish & Remote Deploy Automation** • Developed scripts/create_executive_brief_docx.py generating reports/SITA_Security_Remediation_Executive_Brief.docx and reports/SITA_AI_Role_Advantage_Executive_Brief.docx • Polished homepage UI (frontend/src/pages/HomePage.tsx, home.css) with clean role navigation and status badges • Finalized automated remote deployment scripts (scripts/deploy_remote.py and infrastructure/scripts/deploy.sh)	8
Part 8 Subtotal		**16**

INVOICE SUMMARY

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PART 2 – MULTI-TENANCY HARDENING & GOVERNANCE EVIDENCE (07 – 12 OCT)	**18 hrs**
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PART 6 – G2 AI REPORT GENERATION ENGINE & COPILOT (15 – 18 OCT)	**18 hrs**
PART 7 – AIMI MULTI-PROVIDER GATEWAY & DBSEC AI ENGINE (19 – 21 OCT)	**22 hrs**
PART 8 – C4 ARCHITECTURE VIEWER, DECKS & EXECUTIVE BRIEFS (21 – 22 OCT)	**16 hrs**
TOTAL HOURS	**160 hrs**

BANKING DETAILS

Bank: Bank Zero - Mukuru | **Branch Code:** 435000
Account Number: 70030185715
Account Holder: Bantu InfoTech (Pty) Ltd
Registration No: 2025/228003/07 | **Tax No:** 9637019200
Covered under: Monthly Retainer INV-2026-010 **(R60,000.00)**

Bantu InfoTech (B.iT) • Pretoria, Gauteng
SITA Platform Delivery, FMCG Data Lake Backend Architecture & AIMI AI Integration • October 2026